Perspective-Conditioned Geopolitical Event Prediction

Complete Architecture Specification

Abstract. This document gives the complete confirmed specification of the Lancer architecture. The model runs 100 independent *mini models*, each of which takes four natural-language strings, embeds them through MiniLM-v2 and custom 64-d converters, and produces a binary probability vector $[\hat{p}^+, \hat{p}^-]$. An aggregation head then pools all 100 probability votes and 100 personality vectors into a single final prediction $[\hat{y}^+, \hat{y}^-]$.

Part 1 — The Mini Model (runs 100 times independently)

Each of the 100 mini models receives four raw strings and produces one two-number probability vector. Every mini model has its own independent weights. No parameters are shared between mini models.

Step 1 — String inputs and embedding

The four inputs are plain English strings:

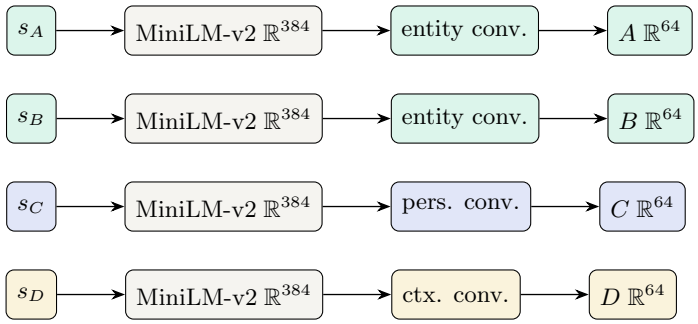
Input	Example	Space
s_A	*"United States"*	entity (shared with B)
s_B	*"Iran"*	entity (shared with A)
s_C	*"realist military analyst"*	personality
s_D	*"Gulf War, 1990, Iraq invades Kuwait"*	context

Each string passes through the two-step embedding pipeline:

$$s \xrightarrow{\text{MiniLM-v2}} \mathbf{z} \in \mathbb{R}^{384} \xrightarrow{\text{converter}_k} \mathbf{e} \in \mathbb{R}^{64}$$

MiniLM-v2 is a frozen pretrained sentence transformer. Each converter_k is a learned two-layer

MLP trained with triplet loss to map the 384-d output into a custom 64-d space. Three converters exist: entity, personality, and context. The entity converter is shared between s_A and s_B, which is what makes cosine similarity between A and B geometrically meaningful.



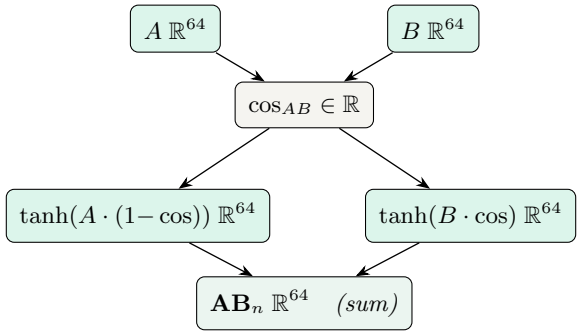
Step 2 — Actor–receiver blend (AB_n)

Cosine similarity measures how alike A and B are in the shared embedding space:

$$\cos_{AB} \;=\; \frac{A \cdot B}{\|A\|\,\|B\|} \;\in\; [-1,\,1]$$

The blend weights each entity by how (dis)similar they are:

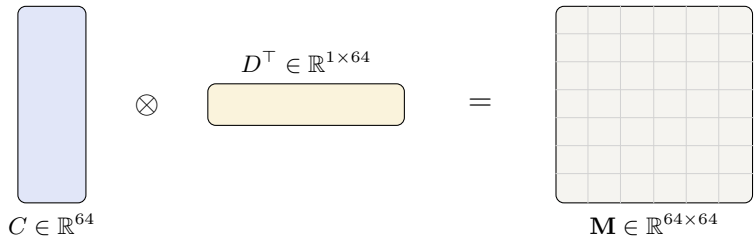
$$\mathrm{AB}_n \;=\; \underbrace{\tanh\!\big(A \cdot (1 - \cos_{AB})\big)}_{\text{actor dominates when } A \neq B} \;+\; \underbrace{\tanh\!\big(B \cdot \cos_{AB}\big)}_{\text{receiver dominates when } A \approx B} \;\in\; \mathbb{R}^{64}$$



Step 3 — Personality–context outer product matrix

C and D are combined via outer product:

$$\mathbf{M} \;=\; C \otimes D \;\in\; \mathbb{R}^{64\times 64} \quad \text{(4096 values per cell)}$$

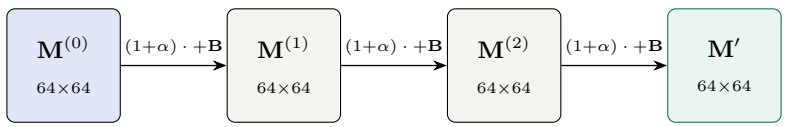


Step 4 — Residual refinement ($\times 3$)

Three passes sharpen $\mathbf{M}$. Each pass applies a shared scalar α and a per-cell bias matrix $\mathbf{B}$:

$$\mathbf{M}^{(t)} \;=\; (1+\alpha)\,\mathbf{M}^{(t-1)} \;+\; \mathbf{B} \qquad t = 1, 2, 3$$

$$\alpha \in \mathbb{R} \text{ shared}, \quad \mathbf{B} \in \mathbb{R}^{64\times 64} \text{ per-cell learned bias}$$

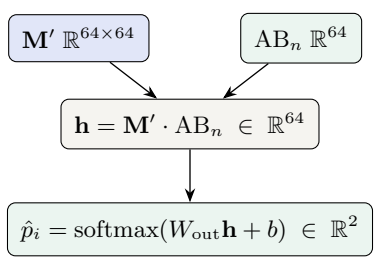


Step 5 — Collapse to binary probability

$\mathbf{M}' \in \mathbb{R}^{64\times 64}$ is multiplied by $\mathrm{AB}_n \in \mathbb{R}^{64}$ to yield a 64-d hidden vector, which is projected and softmax'd to produce the mini model's output:

$$\mathbf{h} \;=\; \mathbf{M}' \cdot \mathrm{AB}_n \;\in\; \mathbb{R}^{64}$$

$$\hat{p}_i \;=\; \mathrm{softmax}\big(W_{\text{out}}^{(i)}\,\mathbf{h} + b_{\text{out}}^{(i)}\big) \;\in\; \mathbb{R}^2 \qquad W_{\text{out}}^{(i)} \in \mathbb{R}^{2\times 64}$$



Mini model output: $\hat{p}_i = [\hat{p}_i^+,\, \hat{p}_i^-] \in \mathbb{R}^2, \quad \hat{p}_i^+ + \hat{p}_i^- = 1$

This runs 100 times independently. Mini model i sees its own $(s_A,\, s_B,\, s_{C_i},\, s_{D_i})$ and produces its own $\hat{p}_i$. No weights are shared between mini models.

Part 2 — Aggregation Head (runs once)

After all 100 mini models have run, the aggregation head combines their probability votes with their personality vectors to produce the final prediction.

Step 6 — Personality sharpening (C_i')

Each personality vector C_i is sharpened and projected:

(a) Scale and softmax:

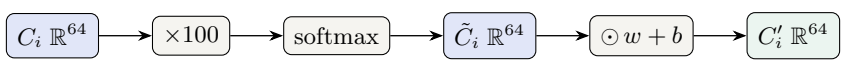
$$\tilde{C}_i \;=\; \text{softmax}(100 \cdot C_i) \;\in\; \mathbb{R}^{64}$$

Multiplying by 100 before softmax amplifies the most activated dimensions so dominant personality traits stand out clearly.

(b) Pairwise learned weights and bias:

$$C_i' \;=\; w \;\odot\; \tilde{C}_i \;+\; b \;\in\; \mathbb{R}^{64}$$

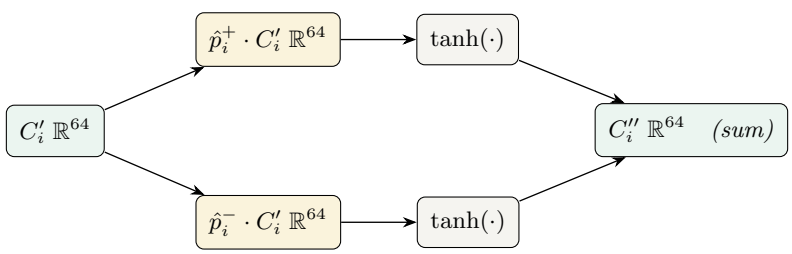
$w, b \in \mathbb{R}^{64}$ are shared learned parameters. $\odot$ is element-wise multiplication.



Step 7 — Confidence gating (C_i'')

Two copies of C_i' are scaled by $\hat{p}_i^+$ and $\hat{p}_i^-$ respectively. Tanh is applied to each and the results are summed:

$$C_i'' \;=\; \tanh\!\big(\hat{p}_i^+ \cdot C_i'\big) \;+\; \tanh\!\big(\hat{p}_i^- \cdot C_i'\big) \;\in\; \mathbb{R}^{64}$$

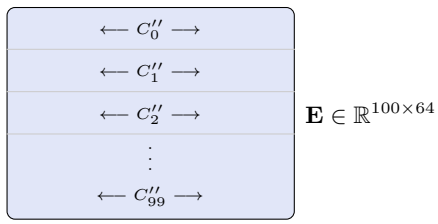


Intuition. A confident mini model (one probability near 1) pushes C_i'' hard toward tanh saturation — a loud, decisive vote. An uncertain mini model (both probabilities near 0.5) produces a soft, muted C_i'' — a quiet, hedged vote. Confidence automatically modulates the volume of each personality's contribution.

Step 8 — Build the ensemble matrix E

All 100 confidence-gated vectors are stacked row-wise:

$$\mathbf{E} \;=\; \begin{pmatrix} \longleftarrow C''_0 \longrightarrow \\ \longleftarrow C''_1 \longrightarrow \\ \vdots \\ \longleftarrow C''_{99} \longrightarrow \end{pmatrix} \;\in\; \mathbb{R}^{100\times 64}$$

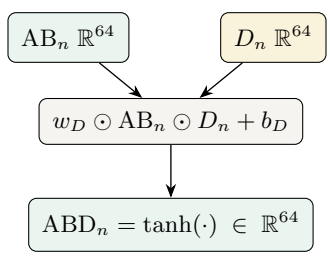


Step 9 — Context-fused situation vector (ABD_n)

The actor-receiver blend AB_n is fused with the context vector D_n through a learned element-wise transform:

$$\mathrm{ABD}_n \;=\; \tanh\Big(w_D \;\odot\; \mathrm{AB}_n \;\odot\; D_n \;+\; b_D\Big) \;\in\; \mathbb{R}^{64}$$

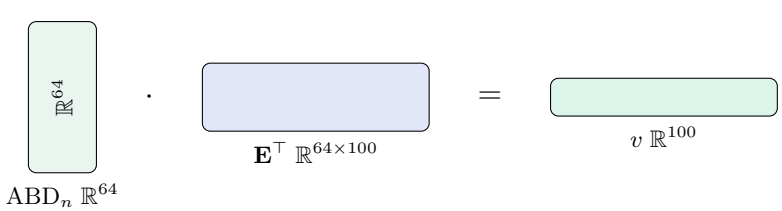
w_D, $b_D \in \mathbb{R}^{64}$ are learned. The result encodes the actor–receiver relationship as seen through the lens of the current global context.



Step 10 — Poll the ensemble

ABD_n is dotted against $\mathbf{E}^\top$ to score every personality's relevance to the current situation:

$$v \;=\; \mathrm{ABD}_n \;\cdot\; \mathbf{E}^\top \;\in\; \mathbb{R}^{100}$$



Element v_i measures how well the current situation aligns with personality i's confident gated view. High v_i means personality i's lens is highly relevant here.

Step 11 — Bottleneck refinement ($\times 3$)

Three expand–contract passes compress and re-expand v, forcing the model to find the most information-dense summary of all 100 votes before making a final decision:

$$v^{(t)} \;=\; W_{\downarrow}^{(t)}\,\mathrm{ReLU}\big(W_{\uparrow}^{(t)}\,v^{(t-1)}\big)$$

$$v^{(t-1)}\ \mathbb{R}^{100} \xrightarrow{W_{\uparrow}\ \mathbb{R}^{128\times 100}} \mathbb{R}^{128} \xrightarrow{\mathrm{ReLU}} \mathbb{R}^{128} \xrightarrow{W_{\downarrow}\ \mathbb{R}^{100\times 128}} v^{(t)}\ \mathbb{R}^{100}$$

Applied three times from $v^{(0)} = v$, yielding $v^{(3)}$. Each pass has its own independent $W_{\uparrow}^{(t)}$ and $W_{\downarrow}^{(t)}$ — no weights shared between passes.

Step 12 — Final output

$$\hat{y} \;=\; \mathrm{softmax}\big(W_{\mathrm{out}}\,v^{(3)} + b_{\mathrm{out}}\big) \;\in\; \Delta^1$$

$$\hat{y} = [\,\hat{y}^{+},\ \hat{y}^{-}\,], \qquad \hat{y}^{+} + \hat{y}^{-} = 1 \qquad W_{\mathrm{out}} \in \mathbb{R}^{2\times 100}$$

$\hat{y}^{+}$ is the probability entity A takes the specified action toward B. $\hat{y}^{-}$ is the probability it does not.

End-to-End Data Flow

Inside each mini model i (runs 100 times independently):

$$(s_A,\, s_B,\, s_{C_i},\, s_{D_i}) \xrightarrow{\text{MiniLM-v2 + converter}} A,\, B,\, C_i,\, D_i \in \mathbb{R}^{64}$$

$$\xrightarrow{\cos_{AB}\ \text{blend}} \mathrm{AB}_n \in \mathbb{R}^{64}$$

$$\xrightarrow{C_i \otimes D_i} \mathbf{M} \in \mathbb{R}^{64\times 64} \xrightarrow{\times 3\ \text{residual}} \mathbf{M}' \in \mathbb{R}^{64\times 64}$$

$$\xrightarrow{\mathbf{M}'\cdot \mathrm{AB}_n} \mathbf{h} \in \mathbb{R}^{64} \xrightarrow{W_{\text{out}},\ \text{softmax}} \hat{p}_i \in \mathbb{R}^2$$

Aggregation head (runs once after all 100 mini models):

$$\{C_i,\, \hat{p}_i\}_{i=0}^{99} \xrightarrow{\times 100,\ \text{softmax},\ \odot w + b} \{C_i'\} \xrightarrow{\text{confidence gate}} \{C_i''\} \xrightarrow{\text{stack}} \mathbf{E} \in \mathbb{R}^{100\times 64}$$

$$\mathrm{AB}_n,\, D_n \xrightarrow{\odot\, w_D + b_D,\ \tanh} \mathrm{ABD}_n \in \mathbb{R}^{64} \xrightarrow{\cdot\, \mathbf{E}^\top} v \in \mathbb{R}^{100}$$

$$\xrightarrow{\times 3\ \text{bottleneck}} v^{(3)} \in \mathbb{R}^{100} \xrightarrow{W_{\text{out}},\ \text{softmax}} \hat{y} \in \Delta^1$$

Notation Reference

Symbol	Meaning
s_A, s_B, s_C, s_D	Raw input strings
$A, B \in \mathbb{R}^{64}$	Actor and receiver embeddings, shared space
$C_i \in \mathbb{R}^{64}$	Personality embedding for mini-model i
$D_i \in \mathbb{R}^{64}$	Context embedding for mini-model i
$\cos_{AB}$	Cosine similarity of A and B
$\mathrm{AB}_n \in \mathbb{R}^{64}$	Blended actor–receiver vector
$\mathbf{M} \in \mathbb{R}^{64\times 64}$	Outer product $C_i \otimes D_i$
$\mathbf{M}'$	$\mathbf{M}$ after 3 residual passes
$\alpha \in \mathbb{R}$	Shared scalar residual weight
$\mathbf{B} \in \mathbb{R}^{64\times 64}$	Per-cell learned bias matrix
$\mathbf{h} \in \mathbb{R}^{64}$	$\mathbf{M}' \cdot \mathrm{AB}_n$, hidden state
$\hat{p}_i \in \mathbb{R}^2$	Mini-model i probability output
$\tilde{C}_i \in \mathbb{R}^{64}$	Sharpened personality (after $\times 100$, softmax)
$C_i' \in \mathbb{R}^{64}$	After pairwise weights and bias
$C_i'' \in \mathbb{R}^{64}$	Confidence-gated personality vector
$\mathbf{E} \in \mathbb{R}^{100\times 64}$	Stacked ensemble matrix
$\mathrm{ABD}_n \in \mathbb{R}^{64}$	Context-fused situation vector
$v \in \mathbb{R}^{100}$	Poll vector
$v^{(3)} \in \mathbb{R}^{100}$	Refined poll after 3 bottleneck passes
$\hat{y} \in \Delta^1$	Final binary probability output
$\mathrm{converter}_k$	MiniLM $\rightarrow$ 64-d learned MLP, space k
$\odot$	Element-wise (Hadamard) product
$\otimes$	Outer product